Understanding Latent Space Constraints in Variational Autoencoders

Abhinav Devi Prasad Dhanunjaya Lokesh Swarup

Indian Institute of Science (IISc), Bengaluru

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Outline

Objective 1: Latent Space Architecture Comparison

Objective 2: To Show Posterior Collapse in Standard VAEs

Objective 3: Latent Space Interpolation

Objective 1: Latent Space Architecture Comparison

Problem Statement

Core Question: How do **continuous** (VAE), **discrete** (VQ-VAE), and **modified-objective** (InfoVAE) latent constraints affect *generative quality* and *reconstruction accuracy*?

Background: VAE & ELBO

Evidence Lower Bound (ELBO)

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]}_{\text{Reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z|x)\|p(z))}_{\text{Regularisation}}$$

reparameterisation: $z = \mu + \sigma \odot \varepsilon,\ \varepsilon \sim \mathcal{N}(0, I)$

- **Standard VAE** — Continuous Gaussian latent space with KL regularisation, encoder outputs (μ, σ) and samples z via reparameterisation.
- **InfoVAE** — Introducing an explicit global divergence penalty to the objective function: $-(\alpha + \lambda - 1)D_{\text{KL}}(q_\phi(z) \parallel p(z))$. This mathematically forces the entire aggregated posterior $q_\phi(z)$ to perfectly match the true prior $p(z)$.
- **VQ-VAE** — replaces the continuous Gaussian latent space with a discrete codebook, the encoder outputs are mapped to the nearest embedding via nearest lookup from codebook.

Methodology: Model Objectives

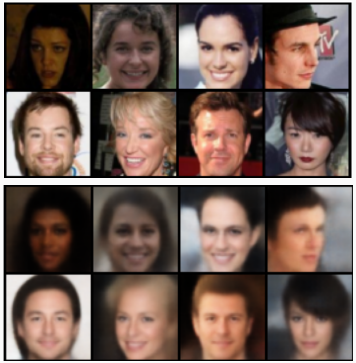
- **Standard VAE**
  - Loss Function:
$$\mathcal{L} = \mathbb{E}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x)\|p(z))$$
  - Encodes inputs into a continuous latent space defined by Gaussian parameters (μ, σ).

- **InfoVAE**
  - **Key Idea:** Replaces KL divergence of aggregated posterior with MMD for easier computation.
  - **Why MMD:** Compares distributions using samples, avoiding complex probability calculations.

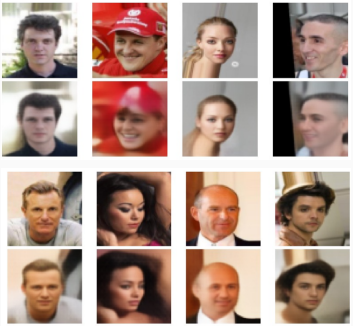
- **VQ-VAE**
  - Loss Function:
$$\mathcal{L} = \|x - \hat{x}\|^2 + \|\mathrm{sg}[z_e] - e\|^2 + \beta\|z_e - \mathrm{sg}[e]\|^2$$
  - Quantization:
$$z_q(x) = \mathrm{sg}\left(\arg\min_{e_k \in \mathcal{E}} \|z_e(x) - e_k\|^2\right)$$

Datasets: CelebA , **Metrics Used:** MSE, FID

Results: Reconstruction Comparison (CelebA)



Standard VAE



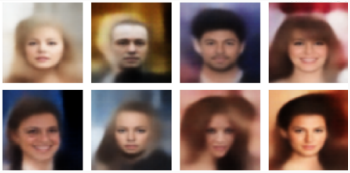
InfoVAE

VQ-VAE

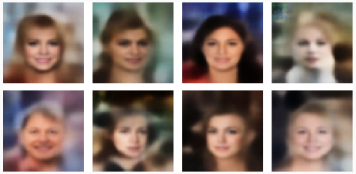
Top: Original , Bottom: Reconstructed

Results: Generated Samples (CelebA)

Generated Samples



Standard VAE



InfoVAE

VQ-VAE

Results: Observations

- **Standard VAE:**
  - *Reconstruction:* Captures basic geometry but suffers from severe, uniform Gaussian blurring.
  - *Generation:* Outputs are structurally weak, overly smoothed, and lack distinct facial features.
- **InfoVAE:**
  - *Reconstruction:* Produces coherent global structures with slightly better color fidelity than the baseline.
  - *Generation:* Avoids total collapse but still exhibits noticeable perceptual blurring and artifacting around edges.
- **VQ-VAE:**
  - *Reconstruction:* Achieves near-perfect structural matching with minimal information loss.
  - *Generation:* Generates the sharpest outputs by far, preserving high-frequency details, distinct edges, and realistic textures.

Results: Quantitative (CelebA)

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)$$

Model	**Reconstruction Loss**	**FID**
Standard VAE	0.0337	85.29
InfoVAE	0.0756	68.21
VQ-VAE	0.0028	18.18

Reconstruction loss measures pixel-level accuracy, while FID evaluates perceptual quality.

Analysis :

- **Standard VAE:** Strict KL penalty forces the approximate posterior $q_\phi(z|x)$ toward $\mathcal{N}(0, I)$. This over-regularization induces a smoothing effect where the decoder averages distinct features, destroying structural coherence.
- **InfoVAE:** Better structure, still blurry. MMD improves latent utilization and preserves global information, but continuous sampling still inherently introduces perceptual blur.
- **VQ-VAE:** A discrete codebook entirely removes Gaussian smoothing. Each input maps deterministically to a single embedding, generating sharp and detailed outputs.

Conclusion: Latent Space Comparison

- **Standard VAE:** Continuous latent space leads to smooth but blurry reconstructions and lower perceptual quality.
- **InfoVAE:** Improves latent utilization and global structure, but still suffers from blur due to continuous sampling.
- **VQ-VAE:** Produces the sharpest and most realistic outputs, with significantly better reconstruction and FID scores.
- **Key Insight:** Latent space structure directly impacts generative quality.
- **Final Takeaway: Discrete latent representations eliminate smoothing and enable high-quality image generation.**

Objective 2: To Show Posterior Collapse in Standard VAEs

Problem Statement

Core Question: How do decoder strength and KL regularization influence latent variable utilization in standard vae, and how do InfoVAE and VQ-VAE mitigate this collapse through modified objectives and architectural changes?

DataSet : CIFAR-10

Background: Posterior Collapse

How Collapse Arises: When the decoder has sufficient capacity to model the data distribution $p(x)$ without relying on the latent space, the optimization process favors minimizing the KL term. This drives $q_\phi(z|x) \approx p(z)$, resulting in latent variables becoming largely independent of the input.

Mitigation Strategies:

- **Reduce Decoder Depth:** Restricts the decoder's strength to prevent it from modeling the data independently.
- **Reduce β in Standard VAE:** Weakens the KL divergence penalty, maintaining non-zero KL and increasing latent usage, but acts as a specific fix that risks a scattered latent space.
- **InfoVAE (Objective-Level Fix)**
- **VQ-VAE (Structural Fix)**

Posterior Collapse: Latent Utilization

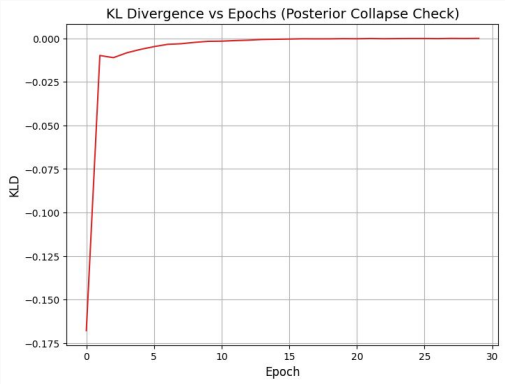
Active Latent Units

Active unit: latent dimension z_i where the variance of its mean across the dataset exceeds $\epsilon = 0.01$.

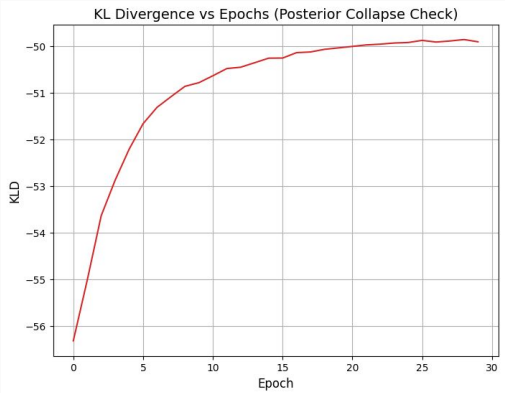
Model Configuration	Active Units (Utilization)
3-Layer Decoder ($\beta = 1.0$)	0 / 32 (0.00%)
5-Layer Decoder ($\beta = 1.0$)	0 / 32 (0.00%)
3-Layer Decoder ($\beta = 0.00025$)	32 / 32 (100.00%)
5-Layer Decoder ($\beta = 0.00025$)	32 / 32 (100.00%)
Info-VAE	128 / 128 (100.00%)
VQ-VAE	512 / 512 (100.00%)

Key Insight: Under strong regularization ($\beta = 1.0$), the decoder completely ignores the latent space (0 active units). Reducing β restores 100% utilization, demonstrating that β controls latent usage while decoder depth has minimal effect on triggering collapse.

Posterior Collapse: Effect of β (CIFAR-10)



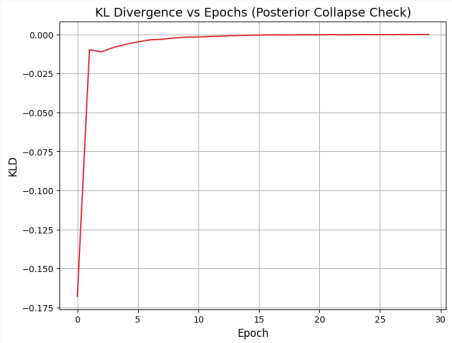
(a) 3-Layer Decoder ($\beta = 1.0$)



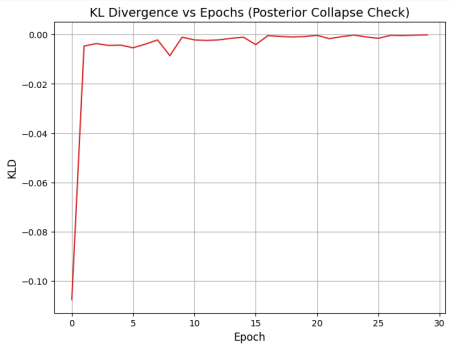
(b) 3-Layer Decoder ($\beta = 0.00025$)

Observation: With standard regularization ($\beta = 1.0$), KL divergence rapidly decreases and stabilizes at a near-constant, very close to zero(≈ 0) value . Reducing β to 0.00025 allows the KL divergence to stabilize at a sustained non-zero value, ensuring latent variables remain informative.

Posterior Collapse: Effect of Decoder Depth



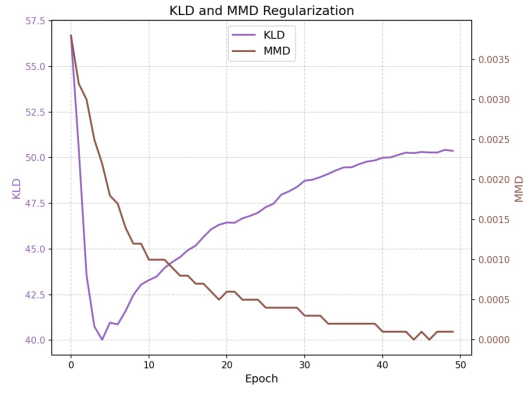
(a) 3-Layer Decoder ($\beta = 1$)



(b) 5-Layer Decoder ($\beta = 1$)

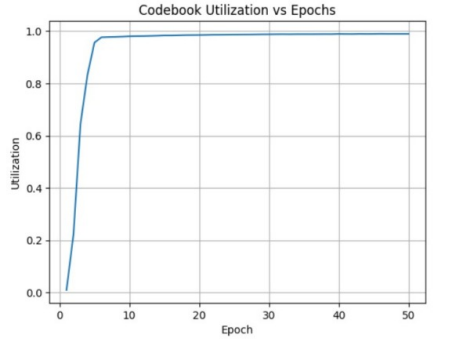
- **Observation:** KL divergence stabilizes at a non-zero value in both configurations.
- **Effect of Depth:** decreasing decoder depth does **not** prevent collapse.

InfoVAE: Absence of Posterior Collapse



- **KLD:** Recovers after an initial drop and increases $\Rightarrow$ latent variables are actively used.
- **MMD:** Decreases and stabilizes near 0 $\Rightarrow$ ensures distribution alignment.
- **Key Insight:** InfoVAE decouples representation learning (KLD) from regularization (MMD).
- **Conclusion:** Avoids posterior collapse while maintaining global alignment.

VQ-VAE: Codebook Utilization



- **Rapid Learning:** Utilization increases sharply during the first few epochs.
- **High Utilization:** Nearly the full codebook is used, showing effective embedding usage.
- **Stable Convergence:** After convergence, utilization remains steady with minimal variation.
- **No Posterior Collapse:** Active codebook usage confirms that latent representations are preserved.

Analysis: Mitigation Strategies Compared

Method	Observed Behaviour & Interpretation
StandardVAE (β-reduction)	Restores non-zero KL divergence (as seen in experiments), ensuring latent usage. However, it weakens regularization and leads to an unstructured latent space.
InfoVAE	KL remains non-zero while MMD decreases, preserving latent usage while keeping the distribution aligned with the prior.
VQ-VAE	Codebook utilization rapidly reaches near 100%, indicating consistent and full latent usage. Discrete quantization structurally prevents collapse by forcing each input to map to a codebook entry.

Conclusion: Posterior Collapse Insights

- **Role of β (kld weight):** Experiments clearly show that β is the primary factor controlling collapse. High β leads to zero KL and inactive latent units, while reducing β restores full utilization.
- **Effect of Decoder Depth:** Increasing decoder depth does **not** prevent collapse, indicating that the issue is not architectural but driven by the objective.
- **InfoVAE (Objective-Level Fix):** Maintains non-zero KL while aligning distributions using MMD, successfully avoiding collapse without weakening regularization.
- **VQ-VAE (Structural Fix):** Discrete codebook ensures consistent latent usage, eliminating collapse by design through enforced quantization.

Key Takeaway: Posterior collapse is fundamentally an objective-driven problem, and robust solutions require either reformulating the objective (InfoVAE) or redesigning the latent structure (VQ-VAE).

Objective 3: Latent Space Interpolation

Problem Statement

Core Question: How smoothly do different models interpolate in latent space, and are the intermediate representations semantically coherent?

Background: Geometry & Interpolation

Linear Interpolation equation

$z_\alpha = (1-\alpha)z_1 + \alpha z_2, \quad \alpha \in [0,1]$

z_1, z_2: latent representations of two input images

Top: Standard VAE Interpolation Middle: InfoVAE Interpolation
Bottom: VQ-VAE Interpolation

Results: Interpolation (CelebA)

$$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \| w_l \odot (\hat{y}^l_{hw} - \hat{y}^l_{0hw}) \|_2^2$$

Model	**LPIPS**
Standard VAE	0.3106
InfoVAE	0.1832
VQ-VAE	0.0155

Lower LPIPS indicates smoother and more consistent interpolation.

Analysis: Latent Space Geometry

- **Continous space (Standard/Info VAE):** Standard VAE forces the latent distribution $q_\phi(z|x)$ toward a standard normal distribution $\mathcal{N}(0, I)$ using KL divergence. This over-regularization forces the model to pack everything into a smooth, continuous blob. When you interpolate halfway between two faces, the decoder essentially mathematically "averages" the distinct features together. This averaging destroys structural coherence, resulting in intermediate images that are blurry, ghost-like, and lack semantic detail.

- **Discrete space:** In VQ-VAE, the encoder's outputs are mapped to the nearest learned embedding via nearest-neighbor lookup. Each of these codebook embeddings represents a highly optimized, concrete structural feature (like a specific edge, texture, or facial structure). Each input maps deterministically to a single embedding, which allows the decoder to generate sharp and detailed outputs.

Conclusion: Interpolation Insights

- **Standard VAE:** Produces continuous but blurry interpolations, with poor semantic consistency.
- **InfoVAE:** Improves smoothness and structure, but still shows instability due to continuous latent sampling.
- **VQ-VAE:** Achieves the smoothest and most coherent interpolation, with clear semantic transitions.
- **Key Insight:** Interpolation quality is directly governed by latent space structure.
- **Final Takeaway: Discrete latent representations enable stable, meaningful, and high-quality interpolation.**

Overall Findings

- **Latent Space Design Matters:** The structure of the latent space directly determines reconstruction quality, generative realism, and interpolation behavior.
- **Continuous Latent Spaces (VAE, InfoVAE):** Lead to smoothing and blur due to Gaussian regularization, limiting perceptual quality despite stable training.
- **Posterior Collapse Insight:** Collapse is primarily driven by the KL weight β, not model architecture. Reducing β restores usage, while InfoVAE avoids collapse through objective reformulation.
- **Discrete Latent Space (VQ-VAE):** Eliminates smoothing and prevents collapse structurally, ensuring consistent and full latent utilization.
- **Overall Outcome: VQ-VAE achieves the best performance across reconstruction, generation, and interpolation.**

Final Takeaway: Redesigning the latent space (discrete representations) is more effective than modifying the objective alone for improving generative models.

Contributions

Name	Contribution
Abhinav	Worked on VQ-VAE, implemented code and experiments; contributed to report writing and prepared the PPT.
Deviprasad	Worked on Standard VAE and VQ-VAE; contributed to report writing and experiments.
Dhanunjaya	Worked on InfoVAE and VQ-VAE; contributed to report writing and experiments.
Lokesh	Studied InfoVAE and Standard VAE; contributed to report writing and experiments.
Swarup	Prepared the PPT, worked on Standard VAE, and contributed to report writing.

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Thank You!